

Sharp Gaussian constants for row deletion in unit-row matrices

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Independently reviewed resolution. The complete proof was checked against the current canonical targets IE-21 and IE-22. The recovered provisional identifiers are confirmed. Authorship is recorded at the submitter's request; substantial AI assistance and reconstruction are disclosed in the submission record. The dated independent agent review records the subsequent checks and supersedes original draft remarks about pending review. This is not external human peer review or formal certification.

Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.

Abstract

Analytic proof drafts give the Gaussian trimmed-second-moment limit for adversarial row retention in independent spherical matrices whenever $n \rightarrow \infty$ and $m/n \rightarrow \infty$, with explicit probability bounds. A projection argument proposes the same sharp universal asymptotic upper constant over all unit-row matrices, without an aspect-ratio requirement for the upper bound. These are analytic results; the accompanying finite verifier does not certify them.

1 Definitions and statements

Fix $0 < \theta < 1$, let $A \in \mathbb{R}^{m \times n}$ have unit Euclidean row norms, and set $k = \lfloor \theta m \rfloor$. Define

$$s_\theta(A) = \min_{|S|=k} \sigma_{\min}(A_S) = \min_{\|x\|_2=1} \left(\sum_{i=1}^k (|Ax|^2)_{(i)} \right)^{1/2},$$

where order is increasing, $\sigma_{\min}$ denotes the variational minimum over unit vectors (zero for a rank-deficient map), and empty sums are zero. The equality follows by interchanging the finite and compact minima.

Let $G \sim N(0, 1)$, $\mathbb{P}(|G| \leq a_\theta) = \theta$, and let φ be its density. Set

$$h_\theta = \mathbb{E}[G^2 \mathbf{1}_{|G| \leq a_\theta}] = \theta - 2a_\theta \varphi(a_\theta), \quad c_\theta = \sqrt{h_\theta}.$$

Theorem 1 (Random rows). *For independent uniform rows on S^{n-1} , along every sequence $n \rightarrow \infty$, $m/n \rightarrow \infty$, in probability*

$$\frac{n}{m} s_\theta(A)^2 \rightarrow h_\theta, \quad \frac{n}{m} \|A\|_2^2 \rightarrow 1, \quad \frac{s_\theta(A)^2}{\|A\|_2^2} \rightarrow h_\theta.$$

Theorem 2 (Universal constant). *Uniformly in all integers $m \geq 1$ and unit-row matrices $A \in \mathbb{R}^{m \times n}$,*

$$\sqrt{n/m} s_\theta(A) \leq c_\theta + o_{n \rightarrow \infty}(1).$$

No smaller constant has the eventual uniform upper-bound property. Along every $n \rightarrow \infty$, $m/n \rightarrow \infty$, the supremum over unit-row matrices of $\sqrt{n/m} s_\theta(A)$ converges to c_θ .

2 Trimming identities

For nonnegative $y_1, \dots, y_m$,

$$\frac{1}{m} \sum_{i=1}^k y_{(i)} = \max_{t \geq 0} \left\{ \frac{k}{m} t - \frac{1}{m} \sum_i (t - y_i)_+ \right\}.$$

The right side is concave piecewise linear, with slope k/m minus the empirical distribution function, and is maximized between the k th and $(k+1)$ st order statistics; use $t = 0$ for $k = 0$.

For a nonnegative random variable Y , define

$$H_\theta(Y) = \inf\{\mathbb{E}(Yw) : 0 \leq w \leq 1, \mathbb{E}w = \theta\}.$$

This is the integral of the lower quantiles through mass θ . For a continuous distribution it is $\mathbb{E}[Y \mathbf{1}_{Y \leq b_\theta}]$. Under any coupling of Y, Z ,

$$|H_\theta(Y) - H_\theta(Z)| \leq \mathbb{E}|Y - Z|.$$

Indeed use the same feasible selectors on the joint space. Allowing selectors to depend on both variables does not change either infimum: conditional expectation given one variable preserves the constraints.

3 Pointwise concentration

Let u be uniform on S^{n-1} , $n \geq 2$, $Y_n = nu_1^2$, and $h_{\theta,n} = H_\theta(Y_n)$. Its law is continuous and its mean is one. Put $L = 2/(1 - \theta)$. For independent copies Y_i , $T_m = m^{-1} \sum_{i=1}^m Y_{(i)}$, and $0 < \epsilon \leq (1 - \theta)/2$, we claim

$$\mathbb{P}(|T_m - h_{\theta,n}| > 2L\epsilon + L/m) \leq 5e^{-2m\epsilon^2}.$$

Let b be the θ -quantile. Since the mean is one, $b \leq 1/(1 - \theta) \leq L$. Hoeffding's bounded-variable inequality gives, outside total probability $4e^{-2m\epsilon^2}$,

$$\left| m^{-1} \sum_i \mathbf{1}_{Y_i \leq b} - \theta \right| \leq \epsilon, \quad \left| m^{-1} \sum_i Y_i \mathbf{1}_{Y_i \leq b} - h_{\theta,n} \right| \leq b\epsilon.$$

Markov's inequality gives $\mathbb{P}(Y_n \leq L) \geq (1 + \theta)/2$. The chance fewer than k samples are at most L is at most $e^{-m(1-\theta)^2/2} \leq e^{-2m\epsilon^2}$. On the complementary events, replacing the sum below b by the k smallest terms changes at most $m\epsilon + 1$ terms, each at most L . This proves the claim. For $k = 0$, use $h_{\theta,n} \leq \theta < 1/m$.

The bounded-variable inequality used here follows from $\mathbb{E}e^{\lambda(X - \mathbb{E}X)} \leq e^{\lambda^2 b^2/8}$ for $0 \leq X \leq b$, independence, and optimization of the Chernoff parameter. The scalar exponential bound follows by convexity on an interval and a one-variable maximization.

4 Covariance and uniform directions

The moments satisfy

$$\mathbb{E}Y_n^r = \frac{n^r(2r-1)!!}{n(n+2) \cdots (n+2r-2)} \leq 2^r r!,$$

so $\mathbb{E}|Y_n - 1|^r \leq 4^r r!$ and, for $|\lambda| \leq 1/8$, $\mathbb{E}e^{\lambda(Y_n - 1)} \leq e^{32\lambda^2}$. The mean-zero linear term vanishes; summing the remaining geometric series gives this bound. Chernoff's inequality then yields, for $0 < u \leq 1$,

$$\mathbb{P}\left(\left|m^{-1} \sum_i Y_i - 1\right| > u\right) \leq 2e^{-mu^2/128}.$$

A $1/4$ -net of the real unit sphere has at most 9^n points; a symmetric matrix's norm is at most twice its maximum absolute quadratic form on that net. For $C = (n/m)A^T A$ this gives

$$\mathbb{P}(\|C - I\|_2 > t) \leq 2 \cdot 9^n e^{-mt^2/512} \quad (0 < t \leq 1).$$

The net cardinality follows by disjoint-ball volume comparison; the quadratic-form estimate follows by approximating a maximizing unit vector.

Let $T(x) = (n/m) \min_{|S|=k} \|A_S x\|_2^2$. On $\|C - I\|_2 \leq t$, every submatrix quadratic form has norm at most $1 + t$, hence

$$|T(x) - T(y)| \leq 2(1 + t) \|x - y\|_2$$

for unit x, y . A δ -net has at most $(1 + 2/\delta)^n$ points. Pointwise concentration, a union bound, and this Lipschitz estimate give

$$\sup_{\|x\|_2=1} |T(x) - h_{\theta,n}| \leq D := 2L\epsilon + L/m + 2(1 + t)\delta$$

except with probability

$$2 \cdot 9^n e^{-mt^2/512} + 5(1 + 2/\delta)^n e^{-2m\epsilon^2},$$

where $0 < t < 1$, $0 < \delta < 1$, $0 < \epsilon \leq (1 - \theta)/2$.

5 Gaussian limit and explicit random error

Couple u with an independent radius $R^2 \sim \chi_n^2$. Then Ru is standard Gaussian, so $G_1^2 = (R^2/n)Y_n$. Independence and the chi-square variance give

$$\mathbb{E}|Y_n - G_1^2| = \mathbb{E}|1 - R^2/n| \leq \sqrt{2/n}, \quad |h_{\theta,n} - h_\theta| \leq \sqrt{2/n}.$$

With the preceding exceptional probability,

$$\left| \frac{n}{m} s_\theta(A)^2 - h_\theta \right| \leq D + \sqrt{2/n},$$

$$\left| \frac{s_\theta(A)^2}{\|A\|_2^2} - h_\theta \right| \leq \frac{D + \sqrt{2/n} + t}{1 - t}.$$

Here $h_\theta \leq 1$. If $Q = m/n \rightarrow \infty$, eventually choose $t = \epsilon = \delta = 32\sqrt{\log Q/Q}$. Parameter restrictions eventually hold; both failure probabilities vanish and the error is

$$O_\theta \left(\sqrt{\frac{n}{m} \log \frac{m}{n}} + n^{-1/2} + m^{-1} \right).$$

No additional condition on how quickly Q diverges is needed. This proves the random theorem.

6 Deterministic projection argument

For arbitrary unit-row A , choose $1 \leq r < n$, set $d = n - r$, and remove the r largest covariance eigenvector directions. In orthonormal coordinates on the remaining subspace, call the projected matrix $B \in \mathbb{R}^{m \times d}$. Since $\text{tr}(A^T A) = m$,

$$\|B\|_2^2 \leq m/(r + 1), \quad \|b_i\|_2 \leq 1, \quad \text{tr}(B^T B) \leq m.$$

Let $g \sim N(0, I_d)$ and $Y_i = (b_i^T g)^2$; these variables may be correlated. Define

$$\Psi_t(g) = \frac{k}{m}t - \frac{1}{m} \sum_i (t - Y_i)_+.$$

Each scalar Gaussian $b_i^T g$ has variance at most one, so

$$\mathbb{E}\Psi_t(g) \leq \theta t - \mathbb{E}(t - G^2)_+ \leq h_\theta.$$

The final inequality is the population trimming formula, maximized at $t = a_\theta^2$.

Gaussian Poincaré gives $\text{Var}(f(g)) \leq \mathbb{E} \|\nabla f(g)\|_2^2$. A proof expands smooth functions in tensor Hermite polynomials: variance sums squared nonconstant coefficients, while the gradient norm weights them by total degree, at least one. Smooth approximation extends the inequality to Lipschitz functions. Since the scalar derivative of $(t - z^2)_+$ is bounded by $2\sqrt{t}$,

$$\text{Var}(\Psi_t(g)) \leq \frac{\|B\|_2^2}{m^2}(4tm) \leq \frac{4t}{r+1}.$$

This handles arbitrary row correlations without asserting independence.

7 Explicit uniform bound and sharpness

For $0 < \delta < 1$, a grid of $[0, L]$ with spacing at most δ has at most $L/\delta + 2$ points. Chebyshev's inequality and the last variance bound show that all grid values obey $\Psi_t \leq h_\theta + \delta$, except with probability at most

$$\frac{4L(L/\delta + 2)}{(r+1)\delta^2}.$$

Since $t \mapsto \Psi_t$ is 1-Lipschitz, its supremum on $[0, L]$ is at most $h_\theta + 2\delta$ on that event's complement.

Also $Z = m^{-1} \|Bg\|_2^2$ has mean at most one and variance $2\text{tr}((B^T B)^2)/m^2 \leq 2/(r+1)$. Thus $Z \leq 2$ except with probability $2/(r+1)$. On that event, the k th order statistic is at most L , because at least $m - k + 1 \geq (1 - \theta)m$ terms are as large. Therefore the full trimming maximum is attained in $[0, L]$; the case $k = 0$ is immediate.

Finally, $\mathbb{P}(\|g\|_2^2 < (1 - \delta)d) \leq 2/(d\delta^2)$. Consequently, if

$$\frac{2}{r+1} + \frac{4L(L/\delta + 2)}{(r+1)\delta^2} + \frac{2}{(n-r)\delta^2} < 1,$$

there exists a realization satisfying all the bounds. Normalize that g to a unit direction in the retained subspace. The trimming identity yields

$$\boxed{\frac{n}{m} s_\theta(A)^2 \leq \frac{n}{(n-r)(1-\delta)} (h_\theta + 2\delta).}$$

All constants are independent of m, A . Taking $r = \lfloor n^{2/3} \rfloor$, $\delta = n^{-1/6}$ makes the condition hold for all sufficiently large n for fixed θ ; its largest term is $O_\theta(n^{-1/6})$. The result is the uniform bound $(n/m) s_\theta(A)^2 \leq h_\theta + O_\theta(n^{-1/6})$.

The random theorem supplies deterministic realizations arbitrarily close to h_θ along every high-aspect-ratio sequence. Thus no smaller constant can satisfy the eventual uniform requirement, and the normalized supremum converges to c_θ . This proves the second theorem's two quantifier claims without interchanging an unjustified random minimum and expectation.

8 Sources and verification limits

Recovered target descriptions: *Sharp row-deletion constant for spherical random matrices*, provisionally IE-21, and *Optimal uniform row-deletion constant*, provisionally IE-22, in OpenProblem-InNLA.¹ Earlier conversational summaries assigned different identifiers to these topics; the present labels are local identifiers only until the repository mapping is checked.

The live source statements could not be retrieved during reconstruction. The precise normalization, retention convention, singular-value definition, and quantifiers in Section 1 are essential. The probability and existence arguments need independent mathematical review; finite tests of the other notes do not certify them.

¹<https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/linear-systems-and-elimination>.